Notes: Card and Krueger (AER, 1994)

Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania

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1 Executive overview

1.1 What the paper asks

How does employment respond to a rise in the minimum wage? The paper studies the 1992 New Jersey minimum-wage increase in a narrowly defined labor market: fast-food restaurants.

1.2 The quasi-experiment

- **Treatment:** New Jersey (NJ) raised its minimum wage from \$4.25 to \$5.05 on April 1, 1992.
- **Control:** Eastern Pennsylvania (PA) remained at the federal minimum wage of \$4.25.
- **Design:** Two waves of a restaurant-level survey:
  - (i) Wave 1: Feb 15–Mar 4, 1992 (pre);
  - (ii) Wave 2: Nov 5–Dec 31, 1992 (post).

1.3 Headline finding

Contrary to the textbook competitive model prediction, the paper finds *no evidence* that the NJ minimum-wage increase reduced employment in fast-food restaurants. If anything, the difference-in-differences estimates point to a *relative increase* in employment in NJ compared to PA.

1.4 Why this design became a template

The paper is an early influential example of a clean *differences-in-differences* (DiD) design:

- same industry and product market,
- geographically close areas,
- pre/post measurement around a policy change,
- explicit robustness checks and placebo tests.

2 Data, measurement, and key constructed variables

2.1 Who is in the sample

Restaurants are from four major fast-food chains: Burger King, KFC, Roy Rogers, and Wendy's (McDonald's is excluded due to low response in earlier pilot work).

The study tracks nearly all wave-1 respondents into wave 2, including closures (with special handling described below).

2.2 Employment measure: full-time equivalent (FTE)

Employment is summarized using full-time equivalents:

$$\text{FTE}_{i,t} \equiv FT_{i,t} + 0.5 \cdot PT_{i,t}, \tag{1}$$

where $FT_{i,t}$ is the number of full-time workers (including managers) and $PT_{i,t}$ is the number of part-time workers. The paper later checks robustness to alternative weights (e.g., 0.4 or 0.6).

2.3 Treatment and timing indicators

Let

$$NJ_i = \mathbf{1}\{\text{restaurant } i \text{ is in New Jersey}\}, \qquad Post_t = \mathbf{1}\{\text{wave 2 (Nov–Dec 1992)}\}.$$

In the paper's main regressions, the outcome is *the change between waves*, so the time indicator is absorbed by first-differencing.

2.4 A key treatment-intensity variable: the "initial wage gap"

The paper constructs a continuous measure of how binding the new law is for each restaurant. Let W_{i1} be the starting wage reported in wave 1. Define the "gap":

$$GAP_i = \begin{cases} 0, & \text{if restaurant } i \text{ is in PA,} \\ 0, & \text{if restaurant } i \text{ is in NJ and } W_{i1} \geq 5.05, \\ \dfrac{5.05 - W_{i1}}{W_{i1}}, & \text{if restaurant } i \text{ is in NJ and } W_{i1} < 5.05. \end{cases} \tag{2}$$

Interpretation: GAP_i is the *proportional wage increase* needed to comply with the new minimum wage. It creates within-NJ variation in "treatment intensity."

3 Core models in the paper

The paper organizes the analysis around first differences (wave 2 minus wave 1) and reduced-form models.

3.1 Model A: raw difference-in-differences (Table 3)

Let $\overline{E}_{s,t}$ be mean employment (FTE) in state $s \in \{NJ, PA\}$ at time $t \in \{1, 2\}$. The simplest DiD estimand is:

$$\widehat{\tau}^{DiD} = \left(\overline{E}_{NJ,2} - \overline{E}_{NJ,1}\right) - \left(\overline{E}_{PA,2} - \overline{E}_{PA,1}\right). \tag{3}$$

The paper also forms an *internal* DiD within NJ by comparing initially low-wage stores to initially high-wage stores.

3.2 Model B: reduced-form first-difference regression (Equation (1a))

Let

$$\Delta E_i \equiv E_{i,2} - E_{i,1}$$

be the change in FTE employment from wave 1 to wave 2. The paper estimates:

$$\Delta E_i = a + b' X_i + c\, NJ_i + \varepsilon_i, \tag{1a}$$

where X_i is a set of observed restaurant characteristics (notably chain dummies, ownership, and sometimes region).

Interpretation. c is a regression-adjusted DiD effect: it compares average employment changes in NJ to PA conditional on X_i.

3.3 Model C: treatment-intensity regression using the wage gap (Equation (1b))

The paper also estimates:

$$\Delta E_i = a' + b' X_i + c'\, GAP_i + \varepsilon_i'. \tag{1b}$$

Interpretation. c' converts treatment intensity into employment effects: it measures how employment changes with the proportional wage increase required by the new minimum wage.

3.4 Model D: proportional employment change (Table 5)

To check that results are not driven by level differences in store size, the paper uses the proportional change:

$$\Delta \widetilde{E}_i \equiv \frac{E_{i,2} - E_{i,1}}{\frac{1}{2}(E_{i,2} + E_{i,1})}, \tag{4}$$

and estimates versions of (1a)–(1b) with $\Delta \widetilde{E}_i$ as the dependent variable.

3.5 Model E: other outcomes (Table 6)

For any outcome Y (hours open, number of cash registers, meal programs, wage profile), the paper estimates:

$$\Delta Y_i = \alpha + \beta' X_i + \gamma\, NJ_i + u_i \qquad \text{or} \qquad \Delta Y_i = \alpha' + \beta' X_i + \gamma'\, GAP_i + u_i', \tag{5}$$

mirroring (1a)–(1b).

3.6 Model F: price effects (Table 7)

Let P_i be the after-tax price of a "full meal" (entree + medium soda + small fries). The paper uses:

$$\Delta \ln P_i = \alpha + \beta' X_i + \gamma\, NJ_i + u_i \qquad \text{or} \qquad \Delta \ln P_i = \alpha' + \beta' X_i + \gamma'\, GAP_i + u_i'. \tag{6}$$

TABLE 1—SAMPLE DESIGN AND RESPONSE RATES

	All	Stores in: NJ	PA
Wave 1, February 15 – March 4, 1992:			
Number of stores in sample frame:[a]	473	364	109
Number of refusals:	63	33	30
Number interviewed:	410	331	79
Response rate (percentage):	86.7	90.9	72.5
Wave 2, November 5 – December 31, 1992:			
Number of stores in sample frame:	410	331	79
Number closed:	6	5	1
Number under rennovation:	2	2	0
Number temporarily closed:[b]	2	2	0
Number of refusals:	1	1	0
Number interviewed:[c]	399	321	78

The NJ dummy identifies the cross-state price differential; the wage gap tests whether more-affected NJ stores raise prices more.

3.7 Model G: store openings at the state level (Table 8)

To address entry/opening concerns beyond the survey of existing stores, the paper analyzes state-level changes in the number of McDonald's restaurants. Let s index states. The main outcomes are:

$$Y_s^{(1)} = \frac{\#\text{stores}_{s,1991} - \#\text{stores}_{s,1986}}{\#\text{stores}_{s,1986}} \quad \text{(proportional growth)}, \tag{7}$$

$$Y_s^{(2)} = \frac{\#\text{new stores}_{s,1986\text{–}1991}}{\#\text{stores}_{s,1986}} \quad \text{(openings rate relative to initial stock)}. \tag{8}$$

They regress these outcomes on alternative measures of the minimum wage in the state plus controls:

$$Y_s = \alpha + \delta \cdot MW_s + \pi_1 \Delta Pop_s + \pi_2 \Delta Unemp_s + \varepsilon_s, \tag{9}$$

weighted by state population.

4 Interpretation of the figure and all tables

This section is written as a "table-by-table" map: what each exhibit is doing and how to read it.

4.1 Table 1: sample design and response rates

Purpose. Establishes the sample frame and attrition across waves.

What it shows.

TABLE 2—MEANS OF KEY VARIABLES

Variable	Stores in: NJ	Stores in: PA	t[a]
1. *Distribution of Store Types (percentages):*			
a. Burger King	41.1	44.3	−0.5
b. KFC	20.5	15.2	1.2
c. Roy Rogers	24.8	21.5	0.6
d. Wendy's	13.6	19.0	−1.1
e. Company-owned	34.1	35.4	−0.2
2. *Means in Wave 1:*			
a. FTE employment	20.4	23.3	−2.0
	(0.51)	(1.35)	
b. Percentage full-time employees	32.8	35.0	−0.7
	(1.3)	(2.7)	
c. Starting wage	4.61	4.63	−0.4
	(0.02)	(0.04)	
d. Wage = $4.25 (percentage)	30.5	32.9	−0.4
	(2.5)	(5.3)	
e. Price of full meal	3.35	3.04	4.0
	(0.04)	(0.07)	
f. Hours open (weekday)	14.4	14.5	−0.3
	(0.2)	(0.3)	
g. Recruiting bonus	23.6	29.1	−1.0
	(2.3)	(5.1)	
3. *Means in Wave 2:*			
a. FTE employment	21.0	21.2	−0.2
	(0.52)	(0.94)	
b. Percentage full-time employees	35.9	30.4	1.8
	(1.4)	(2.8)	
c. Starting wage	5.08	4.62	10.8
	(0.01)	(0.04)	
d. Wage = $4.25 (percentage)	0.0	25.3	—
		(4.9)	
e. Wage = $5.05 (percentage)	85.2	1.3	36.1
	(2.0)	(1.3)	
f. Price of full meal	3.41	3.03	5.0
	(0.04)	(0.07)	
g. Hours open (weekday)	14.4	14.7	−0.8
	(0.2)	(0.3)	
h. Recruiting bonus	20.3	23.4	−0.6
	(2.3)	(4.9)	

- Wave 1: 473 stores in the phone sample frame (364 NJ, 109 PA). Completed interviews: 410 (331 NJ, 79 PA).
- Wave 2: recontact nearly all wave-1 respondents; data exist for essentially the full baseline sample including closure status.

Intuition. High follow-up rates reduce concerns that differential attrition drives the results. The paper also explicitly tracks closures, which helps interpret employment changes as an "overall" effect, not only for surviving stores.

4.2 Table 2: means of key variables

Purpose. Shows baseline comparability and how wages/other variables change from wave 1 to wave 2.

Key reading.

- *Baseline (wave 1):* FTE employment is lower in NJ than PA (NJ stores are smaller on average), while starting wages are very similar across states.
- *After the policy (wave 2):* the NJ starting wage jumps; the fraction of workers at the old $4.25 level drops to (essentially) zero in NJ, while remaining substantial in PA.

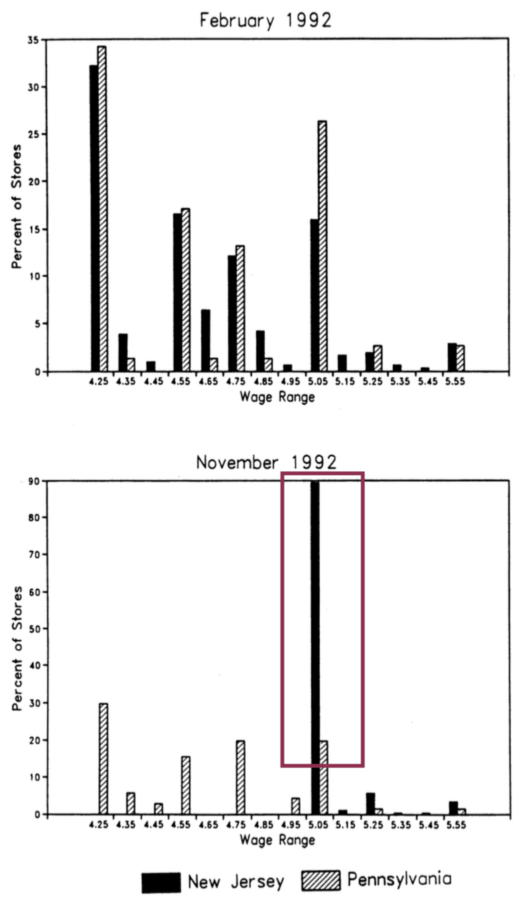

FIGURE 1. DISTRIBUTION OF STARTING WAGE RATES

Interpretation. This table verifies two essential ingredients: (i) the minimum wage is binding in NJ; (ii) there is a valid control group whose wage distribution is not directly affected.

4.3 Figure 1: distribution of starting wage rates

Purpose. Visual confirmation that the policy change creates a large mass point at the new minimum wage.

What to notice.

- February 1992 (pre): NJ and PA wage distributions are similar.
- November 1992 (post): NJ has a sharp spike at $5.05; PA does not.

Intuition. The wage distribution provides a "first-stage" style diagnostic: the treatment meaningfully shifts wages in the treated group, which is required for interpreting any employment effect.

4.4 Table 3: raw DiD in employment (the core result)

Purpose. Computes the DiD effect directly in the employment outcome, with multiple variants:

Table 3—Average Employment Per Store Before and After the Rise in New Jersey Minimum Wage

	Stores by state			Stores in New Jersey[a]			Differences within NJ[b]	
Variable	PA (i)	NJ (ii)	Difference, NJ − PA (iii)	Wage = \$4.25 (iv)	Wage = \$4.26–\$4.99 (v)	Wage ≥ \$5.00 (vi)	Low–high (vii)	Midrange–high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	−2.69 (1.37)	−2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	−2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores[c]	−2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	−2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0[d]	−2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	−2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

- cross-state DiD (NJ vs PA),
- within-NJ comparisons by initial wage category,
- balanced-sample variants and alternative treatment of temporarily closed stores.

Main numbers to remember.

- PA employment falls from 23.33 to 21.17 (change −2.16).
- NJ employment rises from 20.44 to 21.03 (change +0.59).
- DiD: 0.59 − (−2.16) = 2.76 FTE (about 13% of initial NJ employment), with $t \approx 2.0$.

Within-NJ evidence. Stores initially paying \$4.25 increase employment, while initially higher-wage stores contract. This pattern is consistent with the minimum wage being most relevant where it binds.

Robustness within the table. The positive DiD remains when: (i) restricting to the balanced panel of stores with employment data in both waves, (ii) setting employment in temporarily closed stores to zero.

4.5 Table 4: regression-adjusted reduced-form models for employment

Purpose. Implements Equation (1a)/(1b) with covariate adjustment and alternative treatment intensity.

How to read it.

- Columns (i)–(ii): ΔE_i on the NJ dummy.
- Columns (iii)–(v): ΔE_i on the wage gap GAP_i.

Key interpretation.

- The NJ dummy coefficient is about 2.3 FTE, matching the raw DiD in Table 3.
- The wage-gap coefficient is around 15 in the baseline, implying that a store with $GAP = 0.11$ (roughly the mean gap) experiences an employment increase of about $15 \times 0.11 \approx 1.7$ FTE.

TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

Independent variable	Model (i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)	—	—	—
2. Initial wage gap[a]	—	—	15.65 (6.08)	14.92 (6.21)	11.91 (7.39)
3. Controls for chain and ownership[b]	no	yes	no	yes	yes
4. Controls for region[c]	no	no	no	no	yes
5. Standard error of regression	8.79	8.78	8.76	8.76	8.75
6. Probability value for controls[d]	—	0.34	—	0.44	0.40

- Adding region controls attenuates the wage-gap coefficient (and increases its standard error), consistent with measurement error in W_{i1} feeding into GAP_i.

4.6 Table 5: specification tests (robustness and placebo)

Purpose. Stress-tests the reduced-form results across: measurement choices, subsamples, and functional forms.

How to read it. Each row is a different specification. The table reports:

- the NJ dummy coefficient and the wage-gap coefficient for the *level* change model,
- the corresponding coefficients for the *proportional* change model.

What it shows.

- Across many changes (treat temporary closures as closed, exclude managers, reweight part-time, exclude shore towns, add interview-date dummies, drop hard-to-contact stores), the estimated NJ effect remains positive.
- Weighting by initial employment yields a positive and statistically significant proportional effect (NJ dummy ≈ 0.13).
- The placebo test using *only Pennsylvania stores* produces coefficients near zero (and very imprecise), supporting the idea that the wage-gap measure is not mechanically correlated with employment growth absent the policy.

4.7 Table 6: effects on other outcomes (mechanisms and offsets)

Purpose. Tests whether stores adjust margins other than total employment: store operations, nonwage compensation, and wage profiles.

Core message. Most non-employment outcomes show no statistically meaningful NJ-vs-PA shift:

- *Operations:* hours open and cash register measures do not decline in NJ relative to PA.

Table 5—Specification Tests of Reduced-Form Employment Models

Specification	Change in employment NJ dummy (i)	Change in employment Gap measure (ii)	Proportional change in employment NJ dummy (iii)	Proportional change in employment Gap measure (iv)
1. Base specification	2.30 (1.19)	14.92 (6.21)	0.05 (0.05)	0.34 (0.26)
2. Treat four temporarily closed stores as permanently closed[a]	2.20 (1.21)	14.42 (6.31)	0.04 (0.05)	0.34 (0.27)
3. Exclude managers in employment count[b]	2.34 (1.17)	14.69 (6.05)	0.05 (0.07)	0.28 (0.34)
4. Weight part-time as 0.4 × full-time[c]	2.34 (1.20)	15.23 (6.23)	0.06 (0.06)	0.30 (0.33)
5. Weight part-time as 0.6 × full-time[d]	2.27 (1.21)	14.60 (6.26)	0.04 (0.06)	0.17 (0.29)
6. Exclude stores in NJ shore area[e]	2.58 (1.19)	16.88 (6.36)	0.06 (0.05)	0.42 (0.27)
7. Add controls for wave-2 interview date[f]	2.27 (1.20)	15.79 (6.24)	0.05 (0.05)	0.40 (0.26)
8. Exclude stores called more than twice in wave 1[g]	2.41 (1.28)	14.08 (7.11)	0.05 (0.05)	0.31 (0.29)
9. Weight by initial employment[h]	—	—	0.13 (0.05)	0.81 (0.26)
10. Stores in towns around Newark[i]	—	33.75 (16.75)	—	0.90 (0.74)
11. Stores in towns around Camden[j]	—	10.91 (14.09)	—	0.21 (0.70)
12. Pennsylvania stores only[k]	—	−0.30 (22.00)	—	−0.33 (0.74)

Table 6—Effects of Minimum-Wage Increase on Other Outcomes

Outcome measure	Mean change in outcome NJ (i)	PA (ii)	NJ − PA (iii)	Regression of change in outcome variable on: NJ dummy (iv)	Wage gap[a] (v)	Wage gap[b] (vi)
Store Characteristics:						
1. Fraction full-time workers[c] (percentage)	2.64 (1.71)	−4.65 (3.80)	7.29 (4.17)	7.30 (3.96)	33.64 (20.95)	20.28 (24.34)
2. Number of hours open per weekday	−0.00 (0.06)	0.11 (0.08)	−0.11 (0.10)	−0.11 (0.12)	−0.24 (0.65)	0.04 (0.76)
3. Number of cash registers	−0.04 (0.04)	0.13 (0.10)	−0.17 (0.11)	−0.18 (0.10)	−0.31 (0.53)	0.29 (0.62)
4. Number of cash registers open at 11:00 A.M.	−0.03 (0.05)	−0.20 (0.08)	0.17 (0.10)	0.17 (0.12)	0.15 (0.62)	−0.47 (0.74)
Employee Meal Programs:						
5. Low-price meal program (percentage)	−4.67 (2.65)	−1.28 (3.86)	−3.39 (4.68)	−2.01 (5.63)	−30.31 (29.80)	−33.15 (35.04)
6. Free meal program (percentage)	8.41 (2.17)	6.41 (3.33)	2.00 (3.97)	0.49 (4.50)	29.90 (23.75)	36.91 (27.90)
7. Combination of low-price and free meals (percentage)	−4.04 (1.98)	−5.13 (3.11)	1.09 (3.69)	1.20 (4.32)	−11.87 (22.87)	−19.19 (26.81)
Wage Profile:						
8. Time to first raise (weeks)	3.77 (0.89)	1.26 (1.97)	2.51 (2.16)	2.21 (2.03)	4.02 (10.81)	−5.10 (12.74)
9. Usual amount of first raise (cents)	−0.01 (0.01)	−0.02 (0.02)	0.01 (0.02)	0.01 (0.02)	0.03 (0.11)	0.03 (0.11)
10. Slope of wage profile (percent per week)	−0.10 (0.04)	−0.11 (0.09)	0.01 (0.10)	0.01 (0.10)	−0.09 (0.56)	−0.08 (0.57)

TABLE 7—REDUCED-FORM MODELS FOR CHANGE IN THE PRICE OF A FULL MEAL

Independent variable	Dependent variable: change in the log price of a full meal				
	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	0.033 (0.014)	0.037 (0.014)	—	—	—
2. Initial wage gap[a]	—	—	0.077 (0.075)	0.146 (0.074)	0.063 (0.089)
3. Controls for chain and[b] ownership	no	yes	no	yes	yes
4. Controls for region[c]	no	no	no	no	yes
5. Standard error of regression	0.101	0.097	0.102	0.098	0.097

- *Fringe benefits (meal programs):* no evidence that NJ stores systematically reduce free/reduced-price meal programs to offset higher wages.
- *Wage profile:* timing and size of the first raise and the slope of the wage profile do not show significant differential changes.

One notable pattern. The fraction of full-time workers increases in NJ relative to PA in the mean-difference columns. Interpreting this requires caution: it could reflect substitution toward full-time labor, changes in hiring/retention, or measurement noise in hours reporting.

4.8 Table 7: price effects (pass-through)

Purpose. Tests whether NJ restaurants raise prices after the minimum-wage hike, consistent with cost pass-through.

Key fact. The NJ dummy implies that *meal prices rise about 3–4% faster in NJ than PA* between waves (statistically significant).

Interpretation. This supports a cost-based channel: at least part of the wage increase is passed on to consumers. However, within NJ the wage-gap coefficients are not statistically strong, suggesting that price changes are not sharply increasing in measured treatment intensity.

4.9 Table 8: store openings (entry margin)

Purpose. Addresses the concern that minimum wages could discourage entry or store openings, which the two-wave survey of existing restaurants may miss.

How to read it. Using state-level changes in McDonald's restaurants (1986–1991), the minimum-wage variables have *positive* coefficients in most specifications and are not statistically negative.

Interpretation. The evidence is consistent with *small* effects of minimum wages on store openings; it does not support large negative entry effects in this setting, though the authors acknowledge limited power at the state level.

TABLE 8—ESTIMATED EFFECT OF MINIMUM WAGES ON NUMBERS OF MCDONALD'S RESTAURANTS, 1986–1991

Independent variable	Dependent variable: proportional increase in number of stores (i)	(ii)	(iii)	(iv)	Dependent variable: (number of newly opened stores) ÷ (number in 1986) (v)	(vi)	(vii)	(viii)
Minimum-Wage Variable:								
1. Fraction of retail workers in affected wage range 1986[a]	0.33 (0.20)	—	0.13 (0.19)	—	0.37 (0.22)	—	0.16 (0.21)	—
2. (State minimum wage in 1991) ÷ (average retail wage in 1986)[b]	—	0.38 (0.22)	—	0.47 (0.22)	—	0.47 (0.23)	—	0.56 (0.24)
Other Control Variables:								
3. Proportional growth in population, 1986–1991	—	—	0.88 (0.23)	1.03 (0.23)	—	—	0.86 (0.25)	1.04 (0.25)
4. Change in unemployment rates, 1986–1991	—	—	−1.78 (0.62)	−1.40 (0.61)	—	—	−1.85 (0.68)	−1.40 (0.65)
5. Standard error of regression	0.083	0.083	0.071	0.068	0.088	0.088	0.077	0.073

5 How the paper interprets the findings

5.1 Why the results challenge the textbook competitive model

In a perfectly competitive labor market, a binding minimum wage increases labor costs and reduces employment. The paper's central empirical pattern—a positive/zero DiD effect on employment—is hard to reconcile with that baseline model.

5.2 Alternative models consistent with nonnegative employment effects

The paper points toward models where firms have some wage-setting power or where frictions matter:

- **Monopsony/market power in labor markets:** an upward-sloping labor supply to the firm implies that a higher minimum wage can increase employment over some range.
- **Search/wage-posting models:** minimum wages can compress wage dispersion and change equilibrium outcomes.

5.3 A key tension highlighted by the paper

Although some alternative labor-market models can rationalize higher employment, they do not automatically explain the observed *price* increases. This tension motivates thinking about:

- product market structure and pass-through,
- changes in service quality or menu composition,
- heterogeneous adjustment margins (turnover, training, effort) not directly measured here.

6 Takeaways

1. **Design:** NJ vs PA in fast food, two waves around a discrete policy change ⇒ classic DiD.

2. **First stage:** wages in NJ jump to $5.05; PA wages do not (Figure 1, Table 2).

3. **Main result:** employment does not fall in NJ relative to PA; estimated DiD is positive (Table 3) and robust (Tables 4–5).

4. **Margins:** little evidence of systematic reductions in operational scale or nonwage benefits (Table 6).

5. **Prices:** meal prices rise faster in NJ, consistent with partial pass-through (Table 7).

6. **Entry:** no clear evidence of negative effects on store openings (Table 8).

7 Conclusion and intuition

7.1 Coherent summary: what the paper concludes

Card and Krueger (1994) study the 1992 increase in New Jersey's minimum wage (from $4.25 to $5.05) using a two-wave survey of fast-food restaurants and a nearby control group in eastern Pennsylvania. The central conclusion is that, in this setting and time window, the minimum-wage increase *did not reduce employment* in the fast-food sector. If anything, the estimated difference-in-differences (NJ relative to PA, post relative to pre) suggests a *small positive* effect on employment measured in full-time equivalents (FTE).

This result is robust across multiple specifications:

- raw DiD comparisons and regression-adjusted DiD,
- alternative treatments of temporarily closed stores,
- alternative constructions of employment (e.g., different part-time weights),
- models using a continuous "treatment intensity" measure (the initial wage gap to the new minimum),
- proportional employment-change outcomes.

In addition, the paper documents that wages in NJ *did* increase sharply and that prices rose modestly faster in NJ than in PA, consistent with partial pass-through of higher labor costs into product prices rather than large employment cuts.

7.2 What the paper establishes empirically

1. **The "first stage" is strong:** the wage distribution in NJ shifts to the new minimum, while PA does not.

2. **The main outcome is not negative:** employment does not fall in NJ relative to PA after the policy change.

3. **Other margins do not show clear contraction:** there is little evidence of systematic reductions in operational scale (e.g., hours open) or nonwage compensation (e.g., meal benefits) that would substitute for wage cuts.

4. **Prices increase modestly:** meal prices rise faster in NJ, consistent with some cost pass-through.

7.3 Economic intuition: why a higher minimum wage may not reduce employment here

The paper's findings are difficult to reconcile with a frictionless competitive labor market in which firms take wages as given, because that textbook model predicts a binding minimum wage reduces employment.

A coherent intuition consistent with the evidence is that *fast-food labor markets may feature frictions or employer market power*, so that the competitive benchmark is not the right counterfactual. Several mechanisms can rationalize a nonnegative employment effect:

1) Monopsony / upward-sloping labor supply to the firm. If individual firms face an upward-sloping labor supply curve (due to search frictions, mobility costs, or limited information), then firms have wage-setting power and choose wages below the competitive level. In such environments, a higher minimum wage can increase both wages and employment over some range by moving firms closer to the efficient point on the labor supply curve.

2) Reduced turnover and improved hiring/retention. Higher wages can reduce quit rates and improve recruitment, lowering turnover-related costs (training, vacancies, disruptions). Even if the wage bill rises, the effective cost per unit of labor input (after accounting for turnover frictions) may rise much less, and employment need not fall.

3) Adjustment on product-market margins rather than labor quantity. The evidence on prices suggests that some of the cost shock is passed through to consumers. If demand is not extremely elastic and firms have some pricing power, the "incidence" of the policy can be partly on prices, which further weakens the necessity of employment reductions.

7.4 Interpretation and scope: what the conclusion does and does not claim

- The conclusion is local to the episode studied: fast-food restaurants in NJ/PA over a specific post-policy window. It is not automatically a statement about long-run general equilibrium effects or about very large minimum-wage increases.

- The paper's contribution is methodological as well as substantive: it demonstrates how to use a clean DiD design with a nearby control group and intensive robustness checks to evaluate a labor-market policy change.

7.5 Final takeaway

In the NJ fast-food industry in 1992, a sizable minimum-wage hike raised wages but did not reduce employment relative to a nearby control group, suggesting that labor-market frictions or employer market power can overturn the competitive-model prediction in this setting.